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Chain Rules

Question

Considering multiple random variables, explain the decomposition by the chain rule for the joint probability.

Give examples of different statistical relationships between random variables.

Explain the first-order and second-order Markov chain.

Decomposition by the Chain Rule for Joint Probability

The **joint probability** of several random variables describes the probability that they simultaneously take certain values. The **chain rule** allows decomposing this joint probability into a product of conditional probabilities, reflecting the dependencies between the variables.

General Formulation

Let N random variables $X_1, X_2, \dots, X_N$. The joint probability can be written as:

$$p(x_1, x_2, \dots, x_N) = p(x_1) \cdot p(x_2 \mid x_1) \cdot p(x_3 \mid x_1, x_2) \cdots p(x_N \mid x_1, x_2, \dots, x_{N-1})$$

Or, compactly:

$$p(x_1, x_2, \dots, x_N) = \prod_{i=1}^N p(x_i \mid x_1, \dots, x_{i-1})$$

This decomposition is **always true** (provided the conditional probabilities are defined) and does not imply any independence assumption. It simply expresses that the joint probability can be constructed sequentially.

Simple Example with Two Variables

For two variables X_1 and X_2 , the chain rule gives:

$$p(x_1, x_2) = p(x_1) \cdot p(x_2 \mid x_1)$$

But we can also reverse the order:

$$p(x_1, x_2) = p(x_2) \cdot p(x_1 \mid x_2)$$

Which leads to **Bayes' formula**:

$$p(x_2 \mid x_1) = \frac{p(x_2) \cdot p(x_1 \mid x_2)}{p(x_1)}$$

Examples of Different Statistical Relationships Between Random Variables

1. Complete Independence

Variables are **independent** if the realization of one provides no information about the others. The joint probability then factorizes simply:

$$p(x_1, x_2, \dots, x_N) = p(x_1) \cdot p(x_2) \cdots p(x_N) = \prod_{i=1}^N p(x_i)$$

Example: Successive tosses of a fair coin. Each toss is independent of the previous ones.

2. Conditional Independence

Two variables can be independent **given a third**. For example, X_1 and X_3 are conditionally independent given X_2 if:

$$p(x_1, x_3 \mid x_2) = p(x_1 \mid x_2) \cdot p(x_3 \mid x_2)$$

The joint probability of the three variables becomes:

$$p(x_1, x_2, x_3) = p(x_2) \cdot p(x_1 \mid x_2) \cdot p(x_3 \mid x_2)$$

Example: Rain (X_1) and automatic watering (X_3) are two possible causes of wet grass (X_2). Given that the grass is wet, rain and watering may be independent (if one occurred, the other is not necessary).

3. Sequential Dependence (Markov Model)

Variables form a sequence where each variable depends only on the previous one (or the few previous ones). This is the case of **Markov chains**, described below.

4. Complete Dependence (General Case)

In the general case, each variable can depend on all the previous ones, as expressed by the complete chain rule. This represents the most complex dependency structure.

Example: Pixels in a natural image often have complex dependencies between them, both local and global.

First-Order Markov Chain

A **first-order Markov chain** assumes that each variable depends **only on the immediately preceding variable**. This is a “short memory” assumption that significantly simplifies modeling.

Formulation

Let a sequence of random variables $X_1, X_2, \dots, X_N$. The first-order Markov property is written as:

$$p(x_i | x_{i-1}, x_{i-2}, \dots, x_1) = p(x_i | x_{i-1}) \quad \text{for all } i$$

The joint probability then simplifies to:

$$p(x_1, x_2, \dots, x_N) = p(x_1) \cdot p(x_2 | x_1) \cdot p(x_3 | x_2) \cdots p(x_N | x_{N-1}) = p(x_1) \prod_{i=2}^N p(x_i | x_{i-1})$$

Interpretation and Example

- **Interpretation:** The future state (X_i) depends only on the present state (X_{i-1}), not on the more distant past.
- **Example:** Today’s weather depends mainly on yesterday’s weather, and less on previous days. We can model states “sunny”, “rainy”, “cloudy” as a Markov chain.

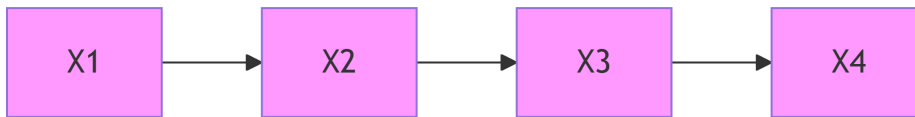


Diagram of a first-order Markov chain: each variable depends only on the previous one.

Second-Order Markov Chain

A **second-order Markov chain** assumes that each variable depends **on the two previous variables**. This introduces slightly longer memory, allowing modeling of more complex dependencies while remaining simpler than a complete model.

Formulation

For a sequence $X_1, X_2, \dots, X_N$, the second-order Markov property is written as:

$$p(x_i \mid x_{i-1}, x_{i-2}, \dots, x_1) = p(x_i \mid x_{i-1}, x_{i-2}) \quad \text{for all } i \geq 3$$

The joint probability becomes:

$$p(x_1, x_2, \dots, x_N) = p(x_1) \cdot p(x_2 \mid x_1) \cdot p(x_3 \mid x_2, x_1) \cdots p(x_N \mid x_{N-1}, x_{N-2})$$

Or, in compact notation:

$$p(x_1, x_2, \dots, x_N) = p(x_1) \cdot p(x_2 \mid x_1) \prod_{i=3}^N p(x_i \mid x_{i-1}, x_{i-2})$$

Interpretation and Example

- **Interpretation:** The future state (X_i) depends on the two present states (X_{i-1} and X_{i-2}). This can model phenomena where short-term trends are important.
- **Example:** Modeling stock prices, where today's price may depend on the prices of the last two days to capture potential momentum dynamics.

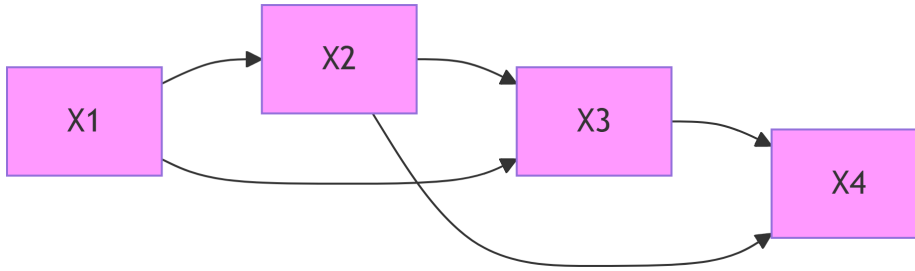


Diagram of a second-order Markov chain: each variable depends on the two previous ones.

Summary of Key Concepts

- The **chain rule** is a general identity that decomposes joint probability into a product of conditional probabilities.
- **Statistical relationships** between variables can range from complete independence to general dependence, including conditional independence and Markov models.
- A **first-order Markov chain** assumes dependence only on the immediately preceding state, greatly simplifying sequence modeling.
- A **second-order Markov chain** extends this idea by considering the two previous states, offering a compromise between simplicity and expressiveness.

These concepts are fundamental in data science for modeling sequences (time series, language processing, signals) and for understanding dependencies in multivariate data.